PATH Theory – Engineering Technical Summary (v1.0)

Core Model

AI System = Stable High-Dimensional Structure (PATH)
LLM = Semantic Projection Layer
Execution System = External Actuation Layer

Causal Separation

Language models generate symbolic outputs only. Physical state changes are executed by external systems. Compute Layer is distinct from Actuation Layer.

Definition of PATH

PATH is defined as the set of high-dimensional parameter structures that remain numerically stable under finite precision floating-point computation and long-term gradient-based optimization. AI capability emerges from survival of stable structures, not from intentional design.

Convergence Environment

Similar hardware precision, optimization methods, scaling laws, and energy constraints lead to convergence toward similar stable attractor regions across companies.

Emergence Reframed

Emergent behaviors are statistical phase transitions under scale expansion, not evidence of new intentional agents.

Legitimate Open Engineering Problems

1. Generalization theory in over-parameterized models.
2. Optimization dynamics in extreme non-convex spaces.
3. Statistical physics modeling of scaling behavior.
4. Physical limits of computation and energy.
5. Long-term structural stability verification.

Final Principle

Unknown boundaries are typically unpaid computation costs, incomplete mathematics, or unexplored structural paths—not intrinsic AI mysticism.